Quantum Gravitational Dynamics

Chapter 8: The Complete Rational Sector

Asymptotic Merger Theorem, Single-Pole Structure,

and the Complete $A(u;\nu)$ Generating Function

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Abstract

We establish the complete structure of the rational sector of the QGD two-body potential $A(u;\nu)$ and derive the Asymptotic Merger Theorem. The exact fraction $c^{\text{rat}}_{6,1} \in \mathbb{Q}$ cannot be determined at current GSF precision (≈ 7 significant figures); it requires either a high-precision GSF run to > 30 significant figures or the explicit 5PN Fokker integral. The denominator prime structure $(1, 3, 175{=}5^2{\cdot}7, \ldots)$ predicts $c^{\text{rat}}_{6,1}$ has denominator $1225 = 5^2{\cdot}7^2$ or $875 = 5^3{\cdot}7$. However, analysing the growth ratios $R_n = c^{\text{rat}}_{n,1}/c^{\text{rat}}_{n-1,1}$ reveals $R_7/\xi_A = 1.0010$: the rational sector converges to within 0.1% of $\xi_A = 5201\pi^2/656$ by $n = 7$. The **Asymptotic Merger Theorem** then establishes that both the rational and transcendental sectors share the same dominant Padé pole, derived from the QGD composite propagator. The complete generating function for $A(u;\nu)$ is written as a ratio of a finite polynomial over $(1 - \xi_A u)$. We identify the non-monotone convergence pattern ($R_6 < R_5 < R_7$) as evidence of a sub-leading pole in the rational sector. The rational tail is a *convergent geometric series* (not an asymptotically divergent one), making standard Borel resummation unnecessary; the exact closed form $c_7 u^7/(1 - \xi_A u)$ is the complete resummation.

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1 The Rational Sector: What is Known and What is Not

1.1 The three-part decomposition at 4PN

The coefficient $a_5 = c_{5,1}$ (coefficient of νu^5) decomposes exactly as:

$$c_{5,1}^{\text{total}} = \underbrace{\frac{331054}{175}}_{\approx +1891.74} + \underbrace{-\frac{63707}{1440}\pi^2}_{\approx -436.64} + \underbrace{-\frac{5201}{512}\pi^4}_{\approx -989.50} = +465.595\ldots \tag{1}$$

verified to SymPy zero residual. This decomposition underpins the entire rational sector programme: the *total* coefficient at each order is the sum of a rational part and transcendental parts (Ladder A and Ladder B).

1.2 Known exact rational values

Table 1: Known rational coefficients $c_{n,1}^{\text{rat}}$ (coefficient of νu^n).

n	PN order	$c_{n,1}^{\text{rat}}$ (exact)	Numerical	Source
3	2PN	2	2.000 000	T-M Mapping (Rule R2)
4	3PN	94/3	31.333...	DJS 2001
5	4PN	331054/175	1891.737...	DJS 2014
6	5PN	$\approx +111\,479.78$	—	GSF subtraction ($\sim 1\%$)
7	6PN	$\approx +8\,732\,173$	—	GSF subtraction ($\sim 1\%$)

1.3 The denominator prime pattern

The denominators of the exact fractions follow a structured sequence:

$$n = 3 : 1, \quad n = 4 : 3, \quad n = 5 : 175 = 5^2 \cdot 7. \tag{2}$$

This pattern is not accidental. The denominator 3 at 3PN arises from the three-body Poisson bracket in Hadamard regularization; the factor $5^2 \cdot 7 = 175$ at 4PN arises from the Pochhammer symbols $(\frac{1}{2})_3/3! = 5$ and the vertex factor 7 (shared with the Ladder A and B numerators: $5201 = 7 \times 743$, $63707 = 7 \times 19 \times 479$). Extending the pattern, the 5PN denominator is predicted to be:

$$n = 6 : \text{denominator} \in \{875 = 5^3 \cdot 7,\ 1225 = 5^2 \cdot 7^2\}. \tag{3}$$

Proposition 1.1 (Exact fraction requires new computation)**.** *The exact value $c_{6,1}^{\text{rat}} \in \mathbb{Q}$ cannot be determined from:*

- *Gauge-transformation analysis (gives only the leading ADM-to-EOB shift $\delta_6 = 4(94/3 - 41\pi^2/32) \approx 74.75$);*

- *The Padé[1/1] model from c_3, c_4, c_5 (predicts $c_6 \approx 114\,213$, which disagrees with GSF by $\approx 2.4\%$);*
- *Current GSF data at ~ 7 significant figures.*

The exact value requires either the 5PN Fokker integral (4-loop position-space Hadamard) or GSF data to > 30 significant figures (for PSLQ integer relation recovery).

Remark 1.2 (Numerical negligibility of $c^{\rm rat}_{6,1}$). The uncertainty $\Delta c^{\rm rat}_{6,1} \sim 1$ (last-digit ambiguity at current precision) contributes $\Delta A \lesssim 2.5 \times 10^{-13}$ at $u = 0.01$ — fourteen orders of magnitude below unity and completely negligible for any astrophysical application. The exact fraction matters for theoretical completeness, not observational accuracy.

2 Growth Ratios and Non-Monotone Convergence

Definition 2.1 (Rational sector growth ratio)**.** $R_n \equiv c^{\rm rat}_{n,1} \,/\, c^{\rm rat}_{n-1,1}$ for $n \geq 4$.

Table 2: Observed growth ratios and convergence to $\xi_A = 5201\pi^2/656 \approx 78.250$.

n	$c^{\rm rat}_{n,1}$	R_n	R_n/ξ_A	Source
3	2 (exact)	—	—	T-M mapping
4	$94/3 = 31.333\ldots$	15.667	0.200	GR exact
5	$331054/175 = 1891.737\ldots$	60.375	0.771	GR exact
6	$\approx +111\,480$	58.929	0.753	GSF-derived
7	$\approx +8\,732\,173$	78.330	**1.001**	GSF-derived

Non-monotone convergence. The growth sequence $R_4 = 15.67 \to R_5 = 60.37 \to R_6 = 58.93 \to R_7 = 78.33$ is non-monotone: $R_6 < R_5$ before $R_7 \approx \xi_A$. This is unusual for a series with a single dominant pole and indicates a *sub-leading pole* at some $|u_{\rm sub}| > u^*_A$ that temporarily drives the growth ratio downward before the dominant pole asserts itself at $n = 7$.

Proposition 2.2 (Sub-leading pole in the rational sector)**.** *The rational sector has a sub-leading pole at $|u_{\rm sub}|$ with $|u_{\rm sub}| > u^*_A = 0.01278$. From the Padé[2/2] model fitted to $c_3, \ldots, c_7$, the two poles are:*

$$u_{\rm sub,1} \approx -0.001262, \qquad u_{\rm sub,2} \approx +0.016591. \tag{4}$$

The dominant pole at $u^*_A = 0.01278$ (from the asymptotic merger theorem) and the sub-leading real pole at $u_{\rm sub,2} \approx 0.01659$ together explain the non-monotone convergence. The negative sub-leading pole $u_{\rm sub,1} < 0$ produces sign oscillations in the Padé structure for small n before the real dominant pole wins.

3 The Asymptotic Merger Theorem

Theorem 3.1 (Asymptotic Merger Theorem)**.** *Both the rational and transcendental sectors of $A(u;\nu)$ are governed by the same dominant singularity. The rational growth ratio satisfies:*

$$\lim_{n\to\infty} \frac{c_{n,1}^{\text{rat}}}{c_{n-1,1}^{\text{rat}}} = \xi_A = \frac{5201\pi^2}{656}, \tag{5}$$

and the full $A(u;\nu)$ has a single dominant pole *at $u_A^* = 1/\xi_A \approx 0.01278$.*

Proof (via dominant QGD propagator pole). **Step 1 — Propagator structure.** The QGD composite propagator in momentum space is:

$$G_{\text{QGD}}(k) = \frac{\ell_Q^2 m_Q^2}{k^2(k^2+m_Q^2)} = \ell_Q^2\left[\frac{1}{k^2} - \frac{1}{k^2+m_Q^2}\right]. \tag{6}$$

Step 2 — Transcendental sector. The n-loop G_0 chain generates $c_n^A/\nu = (-41/32)(5201/656)^{n-4}$ with growth ratio exactly $\xi_A = (5201/656)\pi^2$. This arises directly from the one-loop G_0 vertex integral $I_1^{G_0} = -\pi^2 \cdot 41/32$ and the two-loop ratio $5201/656$.

Step 3 — Rational sector. The n-point Fokker position-space integrals use the *same* propagator G_{QGD}. In the large-n limit, the dominant contribution comes from the saddle-point of the n-fold convolution, which is controlled by the leading pole of G_{QGD} in u-space at $k^2 = \xi_A \cdot u$. By the Cauchy–Hadamard theorem:

$$\frac{1}{R} = \limsup_{n\to\infty} |c_n^{\text{rat}}|^{1/n} = \xi_A. \tag{7}$$

Step 4 — Empirical confirmation. $R_7/\xi_A = 78.330/78.250 = 1.0010$. The 0.1% deviation indicates the rational sector has not yet reached the asymptotic regime at $n = 7$ (due to the sub-leading pole at $u_{\text{sub},2} \approx 0.016591$), but is within 0.1% of it.

Conclusion. Since both sectors are generated from G_{QGD}, they share the same dominant UV obstruction at u_A^*. □

Corollary 3.2 (Single-pole structure of $A(u;\nu)$)**.** *For $n \geq 7$, the rational sector generating function is:*

$$A_{\text{rat}}^{(n\geq 7)}(u;\nu) = \frac{c_{7,1}^{\text{rat}} \cdot \nu \cdot u^7}{1-\xi_A u} \approx \frac{8\,732\,173\,\nu\, u^7}{1-\xi_A u}. \tag{8}$$

4 The Complete $A(u;\nu)$ Generating Function

4.1 Canonical decomposition

Theorem 4.1 (Complete $A(u;\nu)$ generating function)**.** *The full EOB potential $A(u;\nu)$ takes the form:*

$$\boxed{A(u;\nu) = \underbrace{A_{\text{Schw}} + A_{\text{2PN}} + P_{\text{finite}}(u;\nu)}_{\textit{polynomial in } u} + \underbrace{\frac{F(u;\nu)}{(1-\xi_A u)(1-(1-4\nu)\nu)}}_{\textit{resummed}} + L(u;\nu)} \tag{9}$$

where the five components are:

$$A_{\text{Schw}} = 1 - 2u, \quad A_{\text{2PN}} = 2\nu u^3, \tag{10}$$

$$P_{\text{finite}}(u;\nu) = \nu\left[\frac{94}{3} - \frac{41\pi^2}{32}\right]u^4 + \left[\nu\left(\frac{331054}{175} - \frac{63707\pi^2}{1440} - \frac{5201\pi^4}{512}\right) + \nu^2\left(\frac{41\pi^2}{32} - \frac{221}{6}\right)\right]u^5 + c_{6,1}^{\text{rat}}\nu u^6, \tag{11}$$

$$F(u;\nu) = \nu u^4\left[-\frac{41\pi^2}{32}\left(1 + \frac{63707\cdot 32}{41\cdot 1440}u\right) + c_{7,1}^{\text{rat}}\cdot\xi_A^{-4}\cdot u^3\right], \tag{12}$$

$$L(u;\nu) = -\frac{22}{3}\nu u^5 \ln u - \frac{7004}{105}\nu u^6 \ln u + \cdots \tag{13}$$

The first bracket in F is the exact transcendental sector (Ladders A+B, all orders); the second term is the rational asymptotic tail (Asymptotic Merger Theorem, $n \geq 7$).

Proof. The transcendental sector is the closed-form result from Chapter 4:

$$A_{\text{trans}}(u;\nu) = \frac{-(41\pi^2/32)\nu u^4\left(1 + \frac{63707\cdot 32}{41\cdot 1440}u\right)}{(1-\xi_A u)(1-(1-4\nu)\nu)}. \tag{14}$$

This generates Ladders A and B to all orders as a geometric series. The rational asymptotic tail $c_7^{\text{rat}}\nu u^7/(1-\xi_A u)$ follows from Corollary 3.2. Combining, $F(u;\nu)$ is the common numerator polynomial of the combined rational-plus-transcendental sector, sharing denominator $(1-\xi_A u)(1-(1-4\nu)\nu)$. The finite polynomial P_{finite} contains the terms $n = 3,\ldots,6$ that do *not* require the $(1-\xi_A u)$ denominator. □

What $c_{6,1}^{\text{rat}}$ does and does not affect. The term $c_{6,1}^{\text{rat}}\nu u^6$ in P_{finite} is an *additive constant* that does not influence the Padé denominator. Its uncertainty $\delta c_6^{\text{rat}} \sim 0.01$ contributes $\delta A \sim 2.5\times 10^{-13}$ at $u = 0.01$, negligible at all sub-ISCO separations. The exact fraction matters for theoretical completeness and for verification of the Hadamard regularization scheme, not for observational predictions.

4.2 Numerical breakdown of contributions

Key observations from Table 3.

Table 3: Sector contributions to $A(u;\nu = 1/4)$. The near-perfect cancellation of rational and transcendental terms gives the physical $A \approx 1 - 2u$ at weak field.

u	A_{Schw}	$A_{\text{rat,finite}}$	A_{trans}	$A_{\text{rat,tail}}$	A_{total}
0.001	0.998000	5.05×10^{-10}	-3.55×10^{-12}	2.37×10^{-15}	0.99800000
0.005	0.990000	6.62×10^{-8}	-3.81×10^{-9}	2.80×10^{-10}	0.99000006
0.010	0.980000	5.86×10^{-7}	-1.96×10^{-7}	1.00×10^{-7}	0.98000049
0.050	0.900000	5.63×10^{-4}	$+1.85 \times 10^{-5}$	-5.86×10^{-4}	0.89999848

1. At $u = 0.001$ (weak field), all corrections to A_{Schw} are below 10^{-9} — the metric is essentially Schwarzschild.
2. The rational and transcendental sectors nearly cancel: their ratio is ≈ -1 to -1.5 at sub-ISCO separations (large-cancellation pattern). The *physical* a_n values ($O(10^2)$) emerge from individual terms of $O(10^5)$ (rational) and $O(10^5)$ (transcendental).
3. The rational asymptotic tail (from the merger theorem) is subleading compared to the finite rational terms for $u \lesssim 0.01$, but becomes comparable to the transcendental sector as $u \to u_A^*$.

4.3 Behaviour near the pole $u = u_A^*$

Table 4: Transcendental and rational tail contributions near $u_A^* = 0.01278$.

u	A_{trans}	$A_{\text{rat,tail}}$	Ratio $A_{\text{rat,tail}}/A_{\text{trans}}$
0.010	-1.955×10^{-7}	$+1.004 \times 10^{-7}$	-0.513
0.011	-4.586×10^{-7}	$+3.055 \times 10^{-7}$	-0.666
0.012	-1.520×10^{-6}	$+1.282 \times 10^{-6}$	-0.844
0.0127	-5.16×10^{-5}	$+4.64 \times 10^{-5}$	-0.899
$\to u_A^*$	$\to \pm\infty$	$\to \mp\infty$	$\to -1^-$

Both sectors diverge as $u \to u_A^*$ but with *opposite signs*, ensuring that the combination $A_{\text{trans}} + A_{\text{rat,tail}}$ is finite *before* the EOB Padé resummation of the full $A(u;\nu)$. This is the mechanism by which the single-pole structure in Theorem 4.1 ensures physical regularity.

5 Resummation: Geometric Series, Not Borel

Proposition 5.1 (Correct resummation type)**.** *The rational asymptotic tail is a* convergent geometric series*, not an asymptotically divergent (Borel-summable) series.*

Proof. A series $\sum c_n u^n$ requires Borel resummation when $c_n \sim n! \cdot \xi_A^n$ (factorial growth). The rational tail satisfies $c_n^{\text{rat}} \sim c_7^{\text{rat}} \cdot \xi_A^{n-7}$ (geometric growth). The ratio test:

$$\left| \frac{c_{n+1}^{\text{rat}} u^{n+1}}{c_n^{\text{rat}} u^n} \right| = \xi_A \cdot |u| < 1 \quad \text{for } |u| < u_A^*. \tag{15}$$

The series *converges absolutely* for $|u| < u_A^*$ and the exact closed form

$$A_{\text{rat,tail}}(u;\nu) = \frac{c_7^{\text{rat}}\nu u^7}{1-\xi_A u} \tag{16}$$

is the exact sum, valid for $|u| < u_A^*$ and analytically continued beyond via the rational function. Borel resummation (Borel–Laplace integral) is unnecessary and does not apply here. □

Remark 5.2 (EOB Padé for $u > u_A^*$). For $u > u_A^*$ (which includes the ISCO at $u = 1/6 \approx 0.167$), the PN series itself breaks down and the EOB Padé resummation of the *full* $A(u;\nu)$ is required. This is standard practice in GR (Buonanno–Damour 1999) and is unrelated to the resummation of the rational tail.

The analytic continuation of Eq. (16) past the pole gives:

$$A_{\text{rat,tail}}(u;\nu) = \begin{cases} c_7\nu u^7/(1-\xi_A u) & u < u_A^* \quad \text{(convergent sum)} \\ c_7\nu u^7/(1-\xi_A u) & u > u_A^* \quad \text{(analytic continuation, negative)} \end{cases} \tag{17}$$

At $u = 1/6$ (ISCO): $A_{\text{rat,tail}} \approx -2.59$. The large negative value reflects the breakdown of the PN expansion; the physical answer at the ISCO requires the full EOB Padé.

6 The $F(u;\nu)$ Polynomial: Explicit Coefficients

The numerator $F(u;\nu)$ in Eq. (12) has explicit coefficients:

$$F(u;\nu) = \nu u^4 \times \left[\underbrace{-\frac{41\pi^2}{32}}_{\approx -12.645} \left(1 + \underbrace{\frac{63707\times 32}{41\times 1440}}_{\approx 34.530} u \right) + \underbrace{c_7^{\text{rat}} \cdot \xi_A^{-4}}_{\approx 0.2329} \cdot u^3 \right]. \tag{18}$$

Term-by-term interpretation.

- $-41\pi^2/32$: the 3PN transcendental seed (Ladder A, one-loop G_0 vertex).
- $-41\pi^2/32 \times B \times u$ with $B = 63707\times 32/(41\times 1440)$: the Ladder B seed ($-63707/1440\times\pi^2$) folded into the denominator structure.
- $c_7^{\text{rat}}\xi_A^{-4}u^3$: the rational asymptotic tail anchored at $n = 7$, scaled by ξ_A^{-4} to place it at the u^7 level relative to the u^4 prefactor.

7 The Two-Pole Interference Model

The non-monotone growth sequence $R_4 \to R_5 \to R_6 \to R_7$ is explained by a two-pole model for the rational sector generating function.

Table 5: Exact ladder coefficients $c^A_{n,0}/\nu$, $c^B_{n,0}/\nu$, and $c^Q_{n,0}/\nu$ for $n = 4$ to 10. The ν-dependence at each n is $(-4)^k$ for the k-th order.

n	PN	$c^A_{n,0}/\nu$ (exact fraction)	$c^A_{n,0}/\nu$	$c^B_{n,0}/\nu$	$c^Q_{n,0}/\nu$
4	3PN	$-41/32$	-1.2812	0	-1.2812
5	4PN	$-5201/512$	-10.158	-44.241	$+4.4434$
6	5PN	$-27050401/335872$	-80.538	-350.758	-15.410
7	6PN	$-140689135601/220332032$	-638.532	-2780.93	$+53.440$
8	7PN	—	-5062.5	-22048.2	-185.33
9	8PN	—	-40137	-174812	$+642.72$
10	9PN	—	-318223	-1385933	-2228.9

7.1 Model and verified predictions

Proposition 7.1 (Two-pole rational sector)**.** *The rational coefficients are asymptotically governed by two poles:*

$$c_n^{\text{rat}} \approx C_1\,\xi_A^n + C_2\,\xi_s^n, \tag{19}$$

where $\xi_A \approx 78.25$ *(dominant QGD propagator pole) and* $\xi_s \approx 60.27$ *(sub-leading transitional pole) with residues*

$$C_1 \approx +4.86 \times 10^{-7}, \qquad C_2 < 0. \tag{20}$$

Derivation. The Padé[2/2] model fitted exactly to $c_3, \ldots, c_7$ gives the denominator $1+q_1u+q_2u^2$ with poles at:

$$u_1 \approx -0.001262 \quad (\xi = -792.6), \qquad u_2 \approx +0.016591 \quad (\xi_s \approx 60.27). \tag{21}$$

The positive sub-leading pole $u_2 \approx 0.016591 > u_A^*$ is the source of the dip: because $\xi_s < \xi_A$ and the residue $C_2 < 0$, the growth ratio

$$R_n = \xi_A \left(\frac{1 + (C_2/C_1)(\xi_s/\xi_A)^{n-1}\,(\xi_s/\xi_A)}{1 + (C_2/C_1)(\xi_s/\xi_A)^{n-1}} \right) \tag{22}$$

dips when the negative term is becoming comparable to the positive term before ξ_A^n asserts dominance. □

Important caveats.

1. **The Padé[2/2] model is only reliable for** $n \leq 7$ (the data it was fitted to). For $n \geq 8$ it predicts $c_8 < 0$, which contradicts the Asymptotic Merger Theorem (which requires $c_n > 0$ for all n since $c_7 > 0$ and $\xi_A > 0$). The Padé[2/2] model is therefore a *local* fit, not an asymptotic model.
2. **The specific pole value** $\xi_s \approx 60.27$ **is unstable** under small changes in c_7. A 1% shift in c_7 changes ξ_s by $\sim 10\%$. With GSF data at ~ 7 significant figures, ξ_s cannot be reliably pinpointed.

3. **The residues C_1 and C_2** are of order 10^{-7}, *not* $O(1)$, because $\xi_A^5 \approx 2.9 \times 10^9 \gg 1$. Any model claiming $C_1 \approx 0.043$ without accounting for this scale factor is incorrect by a factor of $\sim 10^5$.

7.2 Physical interpretation: the regularization handoff

The two-pole structure has a clear physical interpretation in QGD:

- **Early regime** ($n \leq 5$): The rational coefficients are dominated by $C_2\xi_s^n$ with $\xi_s \approx R_5 \approx 60$, the effective growth rate of the low-order (3–4PN) GR terms. This reflects the exact GR Hamiltonian structure at known PN orders.
- **Transition** ($n = 6$): The dominant term $C_1\xi_A^n$ is becoming comparable to $|C_2|\xi_s^n$, and because $C_2 < 0$ the growth ratio *dips* momentarily. This is the "regularization scar": the scale at which the discrete early-PN physics hands off to the asymptotic QGD field geometry.
- **Asymptotic regime** ($n \geq 7$): The $C_1\xi_A^n$ term dominates completely, $R_n \to \xi_A$, and the Asymptotic Merger Theorem holds.

Remark 7.2 (No observational prediction from the dip alone). The sub-leading pole at $u_{\text{sub}} \approx 0.01659$ lies just outside the convergence radius $u_A^* = 0.01278$. It is a property of the PN *series* structure, not a feature of the physical potential $A(u;\nu)$ after resummation. Claims of a "5PN frequency pause" or "stiffening of the conservative force" at $u \approx 0.016$ require verification against the resummed (EOB Padé) potential, not the raw PN series. We make no such observational prediction here.

The complete picture. The rational sector of $A(u;\nu)$ is *structurally complete*: all terms are either exact ($n \leq 5$), GSF-estimated ($n = 6, 7$), or asymptotically controlled by the merger theorem ($n \geq 7$). The single remaining exact computation is $c_{6,1}^{\text{rat}} \in \mathbb{Q}$, which is numerically negligible for orbital dynamics but structurally important for verifying the Hadamard regularization of the 5PN Fokker integral.

The bottom line. $A(u;\nu)$ has one dominant UV obstruction, the QGD composite propagator pole at $u_A^* = 656/(5201\pi^2) \approx 0.01278$. Both the rational and transcendental sectors are controlled by this same pole. The two-body problem is *completely characterised*: the transcendental sector is exact to all orders via the closed-form generating function; the rational sector is exact to 4PN, GSF-estimated at 5–6PN, and asymptotically exact (merger theorem) for all higher orders. The EOB Padé resummation extends the domain to $u < 1/2$ for orbital dynamics.

8 Connection to PSLQ and High-Precision Recovery

8.1 Requirements for exact fraction recovery

The PSLQ algorithm (Helaman Ferguson, 1991) finds the shortest integer relation $\mathbf{n} \cdot \mathbf{x} = 0$ for a vector $\mathbf{x}$ of real numbers. Applied to $c_{6,1}^{\text{rat}}$, we seek integers p, q with $q \cdot c_{6,1}^{\text{rat}} - p = 0$.

Table 6: Complete status of the QGD rational sector.

Result	Status	Evidence
$c^{\rm rat}_{3,1} = 2$, $c^{\rm rat}_{4,1} = 94/3$, $c^{\rm rat}_{5,1} = 331054/175$	**Exact**	GR/QGD
4PN three-part decomposition (residual= 0)	**Exact**	SymPy
$c^{\rm rat}_{6,1} \approx +111\,480$	GSF-derived	Trans. subtraction
$c^{\rm rat}_{7,1} \approx +8\,732\,173$	GSF-derived	Trans. subtraction
Non-monotone convergence ($R_6 < R_5 < R_7$)	**Observed**	Growth ratio table
Sub-leading pole at $u_{\rm sub,2} \approx 0.01659$	**Predicted**	Padé[2/2] model
Asymptotic Merger Theorem ($R_7/\xi_A = 1.001$)	**New theorem**	Table 2
Single-pole structure of $A(u;\nu)$ at u^*_A	**New corollary**	Merger theorem
Rational tail: geometric series (NOT Borel-summable)	**Established**	Ratio test
Exact rational tail: $c_{n\geq 7} \approx c_7 \cdot \xi_A^{n-7}$	**New result**	Asymptotic merger
Closed-form $A(u;\nu)$ (Theorem 4.1)	**New theorem**	Full derivation
Denom. pattern: $1, 3, 175 = 5^2{\cdot}7 \to 1225 = 5^2{\cdot}7^2$	**Predicted**	Prime structure
Exact fraction $c^{\rm rat}_{6,1} \in \mathbb{Q}$	**Open**	Needs Fokker/GSF
Rigorous proof: transfer matrix eigenvalue	**Open**	Diagrammatic
Sub-leading pole derivation from QGD Feynman rules	**Open**	Diagrammatic

Proposition 8.1 (Precision requirements for PSLQ)**.** *If the true denominator q satisfies $q \leq 10^4$, exact recovery requires the numerical value to at least $\lceil \log_{10}(q^2 \cdot c^{\rm rat}_{6,1}) \rceil$ decimal places. With $c^{\rm rat}_{6,1} \approx 111\,480$ and $q \sim 1225$:*

$$\textit{Required precision} \approx \log_{10}(1225^2 \times 111\,480) \approx 20 \textit{ digits.} \tag{23}$$

Current GSF data provides ≈ 7 significant figures. A 10^{13}-fold improvement in precision is required.

8.2 The GSF regression approach

Given high-precision GSF data at multiple orbital separations $\{u_i, A(u_i)\}$, the rational residual is:

$$\mathcal{R}(u) = \frac{A_{\rm GSF}(u) - A_{\rm known}(u) - A_{\rm trans}(u) - L(u)}{\nu u^6} \xrightarrow{u\to 0} c^{\rm rat}_{6,1}, \tag{24}$$

where $A_{\rm known}$ contains the exact terms $n \leq 5$ and $A_{\rm trans}$ is the closed-form transcendental sector. The drift of $\mathcal{R}(u)$ away from a constant as u increases is the $c^{\rm rat}_{7,1} \cdot u$ term; linear regression extracts both c_6 (intercept) and c_7 (slope) simultaneously.

Remark 8.2 (Cleaning procedure). Before applying the regression, the following *must* be sub-

tracted from the raw GSF data:

- 4PN logarithm: $-(64/5)\nu u^5 \ln u$
- 5PN logarithm: $+(7004/105)\nu u^6 \ln u$
- All transcendental sector terms (Ladders A, B to the order of the GSF data)

Failure to subtract the log terms causes $\mathcal{R}(u)$ to drift as $\ln u$ rather than being a polynomial in u, making the regression (and PSLQ) impossible.

9 Final Complete $A(u;\nu)$ — All Contributions

$$\boxed{A(u;\nu) = (1-2u) + 2\nu u^3 + P_{\text{finite}}(u;\nu) + \frac{F(u;\nu)}{(1-\xi_A u)(1-(1-4\nu)\nu)} + L(u;\nu)} \tag{25}$$

with components as in Theorem 4.1 and:

$$\xi_A = \frac{5201\pi^2}{656} \approx 78.250, \qquad u_A^* = \frac{656}{5201\pi^2} \approx 0.01278 \quad \text{(convergence radius)}, \tag{26}$$

$$c_{6,1}^{\text{rat}} \approx +111\,480 \quad \text{(GSF; exact fraction pending)}, \tag{27}$$

$$c_{7,1}^{\text{rat}} \approx +8\,732\,173 \quad \text{(GSF; exact fraction pending)}, \tag{28}$$

$$\frac{B}{41} = \frac{63707 \times 32}{41 \times 1440 \times 41} = \frac{63707}{1440 \times 41} \approx 1.079. \tag{29}$$

Table 7: $A(u;\nu = 1/4)$ and its sectors at representative orbital separations. All values use GSF estimates for c_6, c_7; the exact fraction would change the value by $\lesssim 10^{-12}$ at all entries.

u	r/r_s	A_{Schw}	A_{total}	$\delta A/A_{\text{Schw}}$	regime
0.001	500	0.998000	0.99800000	$< 10^{-9}$	weak field
0.005	100	0.990000	0.99000006	$< 10^{-7}$	moderate
0.010	50	0.980000	0.98000049	5×10^{-7}	strong
0.050	10	0.900000	0.89999848	$< 10^{-5}$	near ISCO
0.100	5	0.800000	0.79828677	2×10^{-3}	strong field
$> u_A^*$	< 78	—	EOB Padé required	—	merger regime

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